Issue 42 Proof Summary: Consolidated Chapter 2–7 Theorem Progress

Hansen Econometrics Lean Formalization

Branch `codex/issue-42-ch7-unblocked`, compiled May 9, 2026

Abstract

This document is a self-contained mathematical map of the results added in the consolidated issue-42 branch. The branch is large because it collects many small theorem slices that were first developed as separate checkpoints. This summary explains the mathematical surface now present in Lean, records the main public theorem names, and separates completed theorem coverage from remaining gaps. It is not a replacement for the Lean files; it is intended to make the pull request reviewable.

Contents

1 Conventions and Proof Architecture

The formalization keeps three recurring conventions.

- Convergence in probability is written in Lean with `TendstoInMeasure`; convergence in distribution is written with `TendstoInDistribution`.
- Finite-sample ordinary least squares has a base API requiring an invertibility typeclass. Large-sample theorem statements use totalized estimators such as `olsBetaStar` and textbook-facing wrappers such as `olsBetaOrZero`. The star estimator uses the nonsingular inverse and is total; the OrZero wrapper is the reader-facing branch that agrees with ordinary OLS when the Gram matrix is nonsingular.
- Many Chapter 7 theorems are stated at a condition-package layer. The branch adds direct compact and iid sufficient-condition packages for several robust-HC endpoints, but some assumption surfaces remain stronger than the literal textbook assumptions.

The key organizing idea is to prove reusable asymptotic infrastructure once in Chapter 6 and then reuse it in Chapter 7. The Chapter 7 results are mostly composition theorems: consistency plus score CLTs, continuous mapping, Slutsky, matrix inverse continuity, and Gaussian linear-image laws.

2 Chapter 2: Potential Outcomes and CIA

The branch adds a variable-facing potential-outcomes layer for Hansen's causal notation. Given a treatment indicator D, untreated and treated potential outcomes $Y(0), Y(1)$, and covariates X, the Lean API defines:

$$Y = DY(1) + (1 - D)Y(0), \qquad ACE = \mathbb{E}[Y(1) - Y(0)],$$

and a variable-conditioned CATE

$$ACE(X) = \mathbb{E}[Y(1) - Y(0) \mid X].$$

Main declarations:

- `observedOutcome`, `treatmentEffect`, `averageTreatmentEffect`, and `conditionalAverageTreatmentEffectOn`.
- `PotentialOutcomeCIAOn`: the conditional-independence assumption for potential outcomes, treatment, and covariates.
- `PotentialOutcomeCIAOn.toTreatmentMeanIndependentOn`: CIA implies the mean-independence condition needed for regression-style CATE statements.
- `condExpOn_observedOutcome_treatment_covariates_eq_branch_of_CIA`: the observed-outcome regression on (D, X) selects the treated or untreated branch under CIA.
- `conditionalAverageTreatmentEffectOn_treatment_covariates_eq_of_CIA`: conditioning the treatment effect on (D, X) gives the same CATE as conditioning on X alone, almost everywhere.
- `conditionalAverageTreatmentEffectSurface`, `observedRegressionSurface`, and `observedRegressionTreatmentContrastSurface`: thin pointwise notation bridges for $ACE(x)$, $m(d, x)$, and $m(1, x) - m(0, x)$.
- `conditionalAverageTreatmentEffectOn_eq_surface` and `conditionalAverageTreatmentEffectOn_eq_observedRegressionTreatmentContrastSurface`: supplied pointwise versions pull back to the existing variable-facing CATE.

The result is a usable a.e. version of the potential-outcomes and conditional average-treatment-effect story with pointwise notation bridges layered on top of the a.e. conditional-expectation core.

3 Chapter 3: Finite-Sample Least-Squares Algebra

Chapter 3 receives finite-sample algebra useful for later asymptotics and diagnostics.

3.1 Partitioned Regression and FWL

For a column partition $X = [X_1 \ \ X_2]$, the branch formalizes the Frisch–Waugh– Lovell coefficient formulas. The central public theorem is `fromColsBeta_eq_partitionedBetaFormulas`, which packages the two block coefficient identities. The supporting definitions `partitionedLeftBetaFormula` and `partitionedRightBetaFormula` match Hansen's displayed formulas for the two coefficient blocks.

3.2 Projection, Leverage, and Influence

The branch adds leverage and leave-one-out identities. The formal statements include:

- `leaveOneOutBetaDeleted_eq_leaveOneOutBeta`: the literal row-deleted regression agrees with the reduced Gram/cross-product formula.

- `leaveOneOutBeta_eq_olsBeta_sub_invGram_mulVec`: the leave-one-out coefficient differs from the full-sample coefficient by a leverage-weighted correction.
- `leaveOneOutResidual_eq_inv_one_sub_leverage_mul_residual`: the leave-one-out residual is $e_i/(1-h_{ii})$.
- `fitted_sub_leaveOneOutPrediction_eq_leverage_mul_residual`: the prediction change equals the leverage times the leave-one-out residual.
- `maxPredictionInfluence_eq_maxLeveragePredictionErrorInfluence`: the maximum leave-one-out prediction influence has the leverage-residual expression advertised by the diagnostics.

These are deterministic matrix identities. They later feed Chapter 7 leverage and residual-rate statements.

4 Chapter 4: Finite-Sample Covariance Estimators

The branch extends the finite-sample covariance-estimator layer that Chapter 7 uses asymptotically.

4.1 HC Ordering

For leverage values $0 \le h_{ii} < 1$, the branch proves the positive-semidefinite ordering

$$HC0 \preceq HC2 \preceq HC3.$$

The public matrix-level results are `olsHuberWhiteVarianceEstimator_le_HC2_posSemidef` and `olsHuberWhiteHC2VarianceEstimator_le_HC3_posSemidef`.

4.2 HC2 and HC3 Conditional Expectations

The branch records exact conditional-expectation formulas for leverage-adjusted residual squares:

- `condExp_HC2_adjusted_annihilator_row_sq_eq_sigmaSq` and `condExp_HC2_adjusted_annihilator_row_sq_eq_diagonal`.
- `condExp_HC3_adjusted_annihilator_row_sq_eq_sigmaSq_mul_inv` and `condExp_HC3_adjusted_annihilator_row_sq_eq_diagonal`.
- Matrix-valued expectation wrappers for the corresponding HC2 and HC3 covariance estimators.

The branch also adds singleton-cluster bridges showing that singleton clustered covariance estimators reduce to the heteroskedastic sandwich estimators. The cluster score middle matrix and the unadjusted clustered sandwich are proved positive semidefinite via `sum_vecMulVec_posSemidef`, `olsClusteredScoreMiddle_posSemidef`, and `olsClusteredVarianceEstimator_posSemidef`; the adjusted sandwich is PSD whenever its finite-sample scale factor is nonnegative, and `clusterFiniteSampleAdjustment_nonneg` discharges that condition from $k < n$ and $G > 1$. The CR3-style leave-cluster-out layer now has `ClusterIndex`, `clusterDesign`, `clusterCR3Residual`, `clusterLeaveOutBeta_eq_olsBeta_sub_invGram_mulVec_cr3Residual`, `clusterCR3Score`, `olsClusteredCR3VarianceEstimator`, `olsClusteredCR3VarianceEstimator_eq_clusterCR3Score`, `olsClusteredCR3VarianceEstimator_conservative_of_middle`, and `olsClusteredCR3VarianceEstimator_posSemidef`, covering the deterministic equations (4.52), (4.53), and (4.54) at the estimator-definition/named-score/PSD level and reducing CR3 conservativeness to middle-matrix PSD dominance. This tightens the finite-sample Gauss–Markov and covariance-estimator proof layer. The cluster-block layer now also proves the finite partition identity `clusterIndex_sum_eq_sum`, the Gram and cross-product decompositions `sum_clusterGramContribution_eq_gram` and `sum_clusterCrossContribution_eq_cross`, the named-score sandwich rewrite `olsClusteredVarianceEstimator_eq_clusterScore`, and the residual cluster-score normal-equation consequence `sum_clusterScore_eq_zero`. It also names Hansen's clustered covariance middle and sandwich formulas through `clusterCovarianceMiddle` and `olsClusterConditionalVarianceMatrix`, proves PSD inheritance from PSD cluster covariance blocks, and records the block outer-product/score-middle bridges `clusterCovarianceMiddle_outer_eq_scoreMiddle`, `clusterErrorScore`, `olsBeta_linear_decomposition_clusterScores`, `clusterCovarianceMiddle_errorOuter_eq_clusterErrorScoreMiddle`, `condExp_clusterErrorScoreMiddle_entry_eq_clusterCovarianceMiddle`, `condExp_clusterErrorScoreMiddle_eq_clusterCovarianceMiddle`, `condExp_clusterErrorScoreSandwich_eq_olsClusterConditionalVarianceMatrix`, `olsClusteredVarianceEstimator_eq_clusterCovarianceMiddle_residualOuter`, and `olsClusteredCR3VarianceEstimator_eq_clusterCovarianceMiddle_cr3Outer`.

5 Chapter 5: Normal-Regression Moment Face

The exact random-design Kinal threshold is still not proved. The branch does now record the fixed-design finite-moment consequence of the Gaussian coefficient law: `olsBeta_memLp_of_error_gaussian` shows that, once X is fixed and nonsingular and the error vector is Gaussian, the OLS coefficient vector belongs to every finite L^p space. This isolates the remaining Kinal work as the inverse random-Gram tail threshold rather than the Gaussian finite-moment part.

6 Chapter 6: Asymptotic Infrastructure

Chapter 6 is the main infrastructure layer for issue 42. The branch adds or extends large-sample theorem wrappers that Chapter 7 reuses.

6.1 Convergence in Probability and Distribution

The core CMT and WLLN declarations are:

- `tendstoInMeasure_continuous_comp`: global CMT in probability.
- `tendstoInMeasure_continuousAt_const_comp`: CMT at a point for a constant limit.
- `tendstoInDistribution_continuous_comp`: global CMT in distribution.
- `probabilityMeasure_tendsto_map_of_tendsto_of_ae_continuous` and `tendstoInDistribution_ae_continuous_comp`: the textbook a.s.-continuity CMT in distribution, proved by a Portmanteau closed-set push-forward argument.
- `tendstoInMeasure_wlln`: Banach-valued iid WLLN, obtained by upgrading Mathlib's strong law to convergence in measure.
- `tendstoInMeasure_transformed_wlln`: transformed WLLN for sample means of $h(Y_i)$.

The branch also adds product, sum, matrix-vector, matrix-multiplication, matrix inverse, coordinatewise-product, and finite-sum convergence lemmas used in Chapter 7.

6.2 Cramer–Wold and CLTs

The branch formalizes the Cramer–Wold bridge and iid finite-dimensional CLTs:

- `cramerWold_tendstoInDistribution`: convergence of all scalar projections implies vector convergence in distribution.
- `iidScalarCLT_tendstoInDistribution_gaussian`: scalar iid CLT wrapper over Mathlib.
- `iidVectorProjectionCLT_tendstoInDistribution_gaussian_cov`: scalar projection CLT with covariance identified by the vector covariance matrix.
- `iidVectorCLT_tendstoInDistribution_multivariateGaussian`: multivariate iid CLT with limit $N(0, \mathrm{covMat}(Y_0))$.

The branch also introduces projection-level endpoints for heterogeneous and Lindeberg triangular-array CLTs: `MultivariateLindebergCLTConditions`, `multivariateLindebergCLT_tendstoInDistribution`, `HeterogeneousArrayCLTConditions`, and `heterogeneousArrayCLT_tendstoInDistribution`. These are Cramer–Wold endpoints. The scalar Lyapunov-to-Lindeberg algebra now includes `sq_le_even_moment_div_threshold` and `sq_tail_indicator_le_even_moment_div_threshold`, which bound the scalar Lindeberg tail summand by an even higher moment over the threshold power, plus `integral_sq_tail_le_integral_even_moment_div_threshold`, which lifts that bound to expectations. The finite-row and normalized forms are `sum_integral_sq_tail_le_sum_integral_even_moment_div_threshold` and `normalized_sum_integral_sq_tail_le_normalized_even_moment_bound`. The remaining scalar triangular-array probability engine remains outside this branch.

6.3 Delta Method

The new Delta-method module proves the deterministic remainder statement

$$g(x) - g(\theta) - Dg(\theta)(x - \theta) = o(\|x - \theta\|)$$

from Frechet differentiability and packages the distributional conclusion:

$$p_n(\hat{\theta}_n - \theta) \xrightarrow{d} Z \quad \Longrightarrow \quad p_n(g(\hat{\theta}_n) - g(\theta)) \xrightarrow{d} Dg(\theta)Z,$$

assuming the linearization remainder is $o_p(1)$.

Main declarations:

- `deltaMethod_remainder_isLittleO`.
- `deltaMethod_tendstoInDistribution`.
- `smoothFunction_consistency`.
- `smoothFunction_asymptoticNormality_gaussian`.

Concrete examples cover coordinate square, inverse, product, and ratio maps, with derivative matrices and Gaussian linear-image laws: `coordinateSquareDerivativeMatrix_hasLaw_multivariate`

Gaussian_zero, coordinateInvDerivativeMatrix_hasLaw_multivariateGaussian_zero, coordinateProductDerivativeMatrix_hasLaw_multivariateGaussian_zero, and coordinateRatioDerivativeMatrix_hasLaw_multivariateGaussian_zero.

6.4 Stochastic Order

The branch adds a local stochastic-order layer:

- BoundedInProbability: scalar $O_p(1)$.
- BoundedInProbabilityNorm: norm-valued $O_p(1)$.
- moment-to-O_p and moment-to-o_p bridges, including scaled first moments, natural powers, and positive real powers.
- closure under addition, multiplication, deterministic scaling, and "small times bounded is small".

These statements are the workhorse behind Chapter 7 remainder terms, inverse gaps, max-row rates, residual rates, and leverage rates.

6.5 Uniform Integrability, Maximum Bounds, and Weak Moments

The maximum theorem for row norms is formalized on the nonnegative power scale:

$$n^{-1} \max_{i<n} Z_i \xrightarrow{p} 0$$

from uniform integrability of Z_i. In applications $Z_i = \|X_i\|^r$, and a root CMT gives Hansen's displayed maximum-rate form. Main declarations are uniformIntegrable_tail_eLpNorm_one, uniformIntegrable_one_iff_tail_eLpNorm, uniformIntegrable_one_of_identDistrib_memLp, max_norm_scaled_tendstoInMeasure_zero_of_uniformIntegrable_norm_r, and max_norm_scaled_tendstoInMeasure_zero_of_identDistrib_memLp.

For weak convergence and moments, the branch adds:

- bounded-continuous Portmanteau moment wrappers;
- norm-truncation functions and weak-convergence norm-bound transfer;
- limit integrability under eventual first-moment bounds;
- real clipping functions and clipped-identity weak-moment convergence;
- pointwise and integral bounds showing clipping error is controlled by large-tail integrals;
- conversion from Mathlib's UniformIntegrable tail seminorms to real clipping-tail bounds, and the resulting weak-convergence/UI moment theorem.

The main theorem names are TendstoInDistribution.integral_norm_limit_le_of_eventually_integral_norm_bound, realClipBoundedContinuousFunction, TendstoInDistribution.integral_realClip_tendsto, abs_integral_sub_realClip_le_two_mul_integral_tail_abs, and TendstoInDistribution.integral_tendsto_of_uniformIntegrable.

6.6 Sample-Mean Sharpness

The branch adds exact scalar and vector sample-mean variance formulas:

- `iidSampleMean_variance_eq_inv_card_mul`.
- `iidLinearUnbiasedEstimator_variance_ge_sampleMean`.
- `iidLinearUnbiasedEstimator_covMat_sub_sampleMean_posSemidef`.
- `variance_le_of_eq_add_uncorrelated`.
- `covMat_sub_posSemidef_of_eq_add_uncorrelated`.
- `iidEstimator_variance_ge_sampleMean_of_mean_add_uncorrelated`.
- `iidEstimator_covMat_sub_sampleMean_posSemidef_of_mean_add_uncorrelated`.
- `iidSampleMean_covMat_eq_inv_card_smul`.
- `iidSampleMean_covMat_eq_inv_card_smul_of_identDistrib`.

These prove the sample mean attains the n^{-1} variance/covariance scale for iid observations and that no scalar or finite-dimensional scalar-weighted linear unbiased estimator improves on the equal-weight sample mean. The orthogonal residual lemmas also isolate the Hilbert-space variance algebra and compose it with the sample mean for estimators of the form sample mean plus an uncorrelated residual. They are not the full arbitrary-unbiased-estimator lower bound of Hansen Theorem 6.11 because the theorem-specific orthogonality discharge remains missing.

7 Chapter 7: Least-Squares Asymptotics

The Chapter 7 additions assemble the Chapter 6 infrastructure into OLS asymptotic statements.

7.1 Condition Packages

The public condition structures now include:

- `LeastSquaresConsistencyConditions`;
- `ErrorVarianceConsistencyConditions`;
- `ScoreCLTConditions`;
- `RobustCovarianceConsistencyConditions`;
- `HomoskedasticErrorVariance`;
- `RobustFeasibleHCMomentConditions`;
- `IidRobustFeasibleHCMomentConditions`.

The branch adds conversions among these packages and direct compact/iid sufficient-condition endpoints for many theorem statements. These packages are review aids as well as proof infrastructure: they isolate exactly what is being assumed for each theorem surface.

7.2 OLS Consistency and Normality

The branch proves consistency and asymptotic normality for the totalized and ordinary-wrapper estimators. Key statements include:

- `olsBetaStar_consistent` and OrZero/literal wrappers.
- `scoreVector_sampleCrossMoment_tendstoInDistribution_multivariateGaussian`.
- `feasibleScore_tendstoInDistribution_scoreCLT`.
- `olsBetaStar_vector_tendstoInDistribution_scoreCLT`.
- `olsBetaStar_vector_tendstoInDistribution_multivariateGaussian`.
- `olsBetaOrZero_vector_tendstoInDistribution_multivariateGaussian`.
- `scoreProj_olsBetaStar_tendstoInDistribution_gaussian_cov_all`.

Mathematically, these statements show

$$\sqrt{n}(\hat{\beta}_n - \beta) \xrightarrow{d} N(0, Q^{-1}\Omega Q^{-1})$$

at the current condition layer, with projection-family and vector forms.

7.3 Residual Variance and Homoskedastic Covariance

For Hansen Theorems 7.4 and 7.5, the branch proves consistency of totalized residual-variance and homoskedastic plug-in covariance estimators:

- `olsSigmaSqHatStar_tendstoInMeasure_errorVariance`.
- `olsS2Star_tendstoInMeasure_errorVariance`.
- `olsHomoCovStar_tendstoInMeasure`.
- direct iid endpoints under `IidRobustFeasibleHCMomentConditions`.

7.4 HC0, HC1, HC2, and HC3 Covariance Consistency

For Theorems 7.6 and 7.7, the branch adds a substantial robust-HC consistency layer:

- HC0 middle-matrix and sandwich consistency for feasible residuals.
- HC1 consistency using the finite-sample scale factor `hc1FiniteSampleScale_tendsto_one`.
- HC2 and HC3 consistency through a generic leverage-adjusted middle matrix and max-leverage control.
- compact moment and iid joint-observation endpoints for HC0–HC3.

Representative theorem names are `olsHetCovStar_tendstoInMeasure_of_robustFeasibleHCMomentConditions`, `olsHetCovHC1Star_tendstoInMeasure_of_robustFeasibleHCMomentConditions`, `olsHetCovHC2Star_tendstoInMeasure_of_robustFeasibleHCMomentConditions`, `olsHetCovHC3Star_tendstoInMeasure_of_robustFeasibleHCMomentConditions`, and the corresponding `..._of_iidRobustFeasibleHCMomentConditions` endpoints.

7.5 Functions of Parameters

For Theorems 7.8–7.10, the branch proves both fixed-linear and nonlinear function-of-parameter results:

- continuous-map consistency for $r(\hat{\beta})$;
- nonlinear Delta-method Gaussian limits for totalized and OrZero OLS;
- fixed-linear and random-linear covariance CMTs;
- plug-in derivative covariance consistency for nonlinear functions.

Representative names include `nonlinearFunction_olsBetaStar_delta_tendstoInDistribution_gaussian`, `nonlinearFunction_olsBetaOrZero_delta_tendstoInDistribution_gaussian`, `linMapCov_tendstoInMeasure`, `randomLinearMapCovariance_tendstoInMeasure`, `nonlinearDerivativeCovariance_olsBetaStar_tendstoInMeasure`, and `nonlinearDerivativeCovariance_olsBetaOrZero_tendstoInMeasure`.

7.6 Standard Errors, t Statistics, Confidence Intervals, and Wald Tests

For Theorems 7.11–7.14, the branch adds the inference layer:

- standard-error consistency by continuous mapping;
- studentized scalar limits;
- symmetric confidence-interval coverage from absolute t-statistic convergence;
- multivariate Wald statistic convergence;
- homoskedastic and heteroskedastic HC0–HC3 versions;
- direct compact and iid endpoints where the required condition packages can be discharged.

Core declarations include `studentizedLimit_tendstoInDistribution`, `linMapCovStdError_tendstoInMeasure`, `symmetricCI_coverage_of_abs_tstat_nonpos_tendsto_zero`, `symmetricCI_coverage_of_abs_tstat_standardNormal_se_tendsto_pos`, `waldQuadForm_tendstoInDistribution_of_vector_and_covariance`, `hasLaw_gaussian_mahalanobis_chiSquared`, and the HC0–HC3 scalar and multivariate wrappers in `Inference.lean`, `Normality.lean`, and `SandwichAssembly.lean`.

7.7 Residual Uniformity and Leverage Rates

For Theorems 7.16 and 7.17, the branch proves deterministic inequalities and probabilistic rate wrappers:

- the max residual error is bounded by a coefficient-error factor times a max-row-norm factor;
- the coefficient-error factor is $O_p(1)$ by the vector OLS CLT;
- the max-row-norm factor is $o_p(1)$ by the Chapter 6 maximum theorem;
- max leverage is controlled by inverse-Gram bounds and max row norms.

Representative theorem names are `sqrt_smul_olsBetaStar_sub_boundedInProbabilityNorm`, `sqrt_scaledMaxRowNorm_sq_tendstoInMeasure_zero_of_uniformIntegrable_norm_sq`, `maxResidualErrorStar_tendstoInMeasure_zero_of_uniformIntegrable_rowNorm_sq`, `maxResidualErrorStar_tendstoInMeasure_zero_of_identDistrib_memLp_rowNorm_sq`, `scaledMaxLeverageStar_tendstoInMeasure_zero_of_scaled_maxRowNorm_sq`, `maxLeverageStar_tendstoInMeasure_zero_of_uniformIntegrable_rowNorm_sq`, and `maxLeverageStar_tendstoInMeasure_zero_of_identDistrib_memLp_rowNorm_sq`. The direct iid feasible-HC package derives the squared-row L^1 input for these wrappers from its fourth-row-moment assumption. The branch also adds `IidAssumption72FourthMomentConditions`, which derives the squared-error moment and true-error score outer-product moment from structural-error and regressor fourth moments, then converts to `IidRobustFeasibleHCMomentConditions`. It further adds `IidAssumption72ResponseMomentConditions`, which derives the structural-error fourth moment from the response fourth moment, the linear-model identity, and the regressor fourth-row moment before reusing the fourth-moment package. The combined robust feasible-HC package and the iid package now also derive the feasible-HC third absolute row-weight moment $\mathbb{E}\{|e_0|\|X_0\|^3\} < \infty$ from the score outer-product moment and the fourth-row moment, rather than carrying it as a separate primitive field in those public packages.

7.8 Edgeworth Interface

The branch does not prove Hansen's concrete high-moment cumulant expansion. It does add generic interfaces and the polynomial-shape layer that future work can instantiate:

- `FirstOrderEdgeworthExpansion`;
- `SecondOrderEdgeworthExpansion`;
- scaled-remainder consequences;
- unscaled second-order approximation-error consequences;
- CDF convergence consequences;
- a bridge from the second-order interface to the first-order interface;
- `edgeworthP1Polynomial` and `edgeworthP2Polynomial`, recording the even quadratic and odd degree-five textbook shapes;
- `SecondOrderEdgeworthExpansion.symmetric_interval_scaled_remainder_tendsto_zero`, proving the two-sided cancellation of the $n^{-1/2}$ Edgeworth term.
- `SecondOrderEdgeworthExpansion.symmetric_interval_scaled_remainder_tendsto_zero_polynomial`, the coefficient-specialized version for the Hansen polynomial forms.

Thus Theorem 7.15 has a reusable formal target shape and the central-interval cancellation consequence, but not the high-moment cumulant coefficient proof.

8 Remaining Gaps

The branch is not a claim that Chapters 2–7 are literally complete. The main remaining gaps are:

Area	Remaining work

Chapter 4	Clustered covariance asymptotics beyond the finite cluster-block identities; CR3 middle-moment comparison needed for the full conditional conservativeness theorem.
Chapter 5	Exact random-design Kinal theorem 5.5 inverse-Gram moment threshold.
Chapter 6	Scalar triangular-array Lindeberg/Lyapunov sufficient-condition probability-engine discharge for Theorems 6.4 and 6.5 beyond the deterministic and expected finite-row even-moment tail inequalities; arbitrary-unbiased-estimator lower bound for Theorem 6.11.
Chapter 7	Further tightening of the remaining score-CLT public package toward literal textbook iid assumptions; concrete high-moment cumulant expansion for Theorem 7.15.

9 Files Touched by the Consolidated Branch

The main proof files changed by the branch are:

- `HansenEconometrics/AsymptoticUtils.lean`
- `HansenEconometrics/AsymptoticUtils/DeltaMethod.lean`
- `HansenEconometrics/AsymptoticUtils/MaxBounds.lean`
- `HansenEconometrics/AsymptoticUtils/StochasticOrder.lean`
- `HansenEconometrics/Chapter2PotentialOutcomes.lean`
- `HansenEconometrics/Chapter3FWL.lean`
- `HansenEconometrics/Chapter3LeastSquaresAlgebra.lean`
- `HansenEconometrics/Chapter3Projections.lean`
- `HansenEconometrics/Chapter4LeastSquaresRegression.lean`
- `HansenEconometrics/Chapter6Asymptotics.lean`
- `HansenEconometrics/Chapter7Asymptotics/Consistency.lean`
- `HansenEconometrics/Chapter7Asymptotics/Inference.lean`
- `HansenEconometrics/Chapter7Asymptotics/MiddleConsistency.lean`
- `HansenEconometrics/Chapter7Asymptotics/Normality.lean`
- `HansenEconometrics/Chapter7Asymptotics/SampleMiddle.lean`
- `HansenEconometrics/Chapter7Asymptotics/SandwichAssembly.lean`
- `HansenEconometrics/ProbabilityUtils.lean`

The chapter inventories were updated alongside the Lean files, and this document is intended as a single reviewable guide to the mathematical content.